

Reviewer Pack — Noncommutative Norm Gap in Read-Twice Branching Programs for...

Entry aed3c06488a0 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Noncommutative Norm Gap in Read-Twice Branching Programs for IP₂

Entry ID: aed3c06488a0 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-08 18:36:26 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Read-twice Branching Programs (BP) for IP₂

Statement:

For a read-twice BP P computing IP₂ on n variables, the noncommutative operator norm $\|T_P\|_\infty$ of its transition matrix T_P satisfies $\|T_P\|_\infty = O(\log n)$, whereas for a general BP computing IP₂, $\|T_P\|_\infty = \Omega(n)$.

Rationale (proposer's reasoning):

Noncommutative operator norms capture the structure of the transition matrix in a way that reflects the read-twice constraint, potentially exposing a gap between structured and unrestricted BPs. This connects algebraic topology (via operator algebras) to branching program complexity.

Taxonomy category: BP_READTWICE (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 7fc2026e3491c975

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from typing import List, Dict

def generate_random_matrix(n: int) -> List[List[float]]:
    return [[random.random() for _ in range(n)] for _ in range(n)]

def matrix_multiply(A: List[List[float]], B: List[List[float]]) -> List[List[float]]:
    n = len(A)
    C = [[0.0 for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def spectral_norm(matrix: List[List[float]]) -> float:
    n = len(matrix)
    v = [1.0 / math.sqrt(n)] * n
    for _ in range(100): # Power iteration method
        v = matrix_multiply(matrix, v)
        norm = sum(x**2 for x in v) ** 0.5
        v = [x / norm for x in v]
    return max(abs(x) for x in v)

def run_trial(seed: int) -> Dict[str, any]:
    random.seed(seed)
    n = random.randint(5, 40)

    # Generate a read-twice branching program matrix
    read_twice_matrix = generate_random_matrix(n)
    read_twice_norm = spectral_norm(read_twice_matrix)

    # Generate a general branching program matrix
    general_matrix = generate_random_matrix(n)
    general_norm = spectral_norm(general_matrix)

    return {
        "metric_name": "Noncommutative Operator Norm",
```

```

        "metric_value": read_twice_norm / general_norm,
        "instances_tested": 1,
        "conjecture_holds": read_twice_norm <= math.log(n),
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = (sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))**0.5
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 65, in <module>
    result = run_trial(seed)
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 45, in run_trial
    read_twice_norm = spectral_norm(read_twice_matrix)
                          ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 34, in spectral_norm
    v = matrix_multiply(matrix, v)
        ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 27, in matrix_multiply
    C[i][j] += A[i][k] * B[k][j]
               ~~~~^^^
TypeError: 'float' object is not subscriptable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with `TypeError`, preventing data collection. The error suggests a code issue rather than a mathematical failure. | next: Fix the `TypeError` in `matrix_multiply` by ensuring proper handling of matrix and vector types

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	113327
2	propose	ollama_remote	qwen3:8b	0	0	88189
3	preregistration	ollama_remote	qwen3:8b	0	0	24901
4	novelty	ollama_remote	qwen3:8b	0	0	20718
5	novelty	ollama_remote	qwen3:8b	0	0	13208
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11043
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7812
8	judge	ollama_remote	qwen3:8b	0	0	16469

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 295666 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/aed3c06488a0.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/aed3c06488a0.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/aed3c06488a0.tar.gz` (if generated)